Chapter 8

Answer 8.1

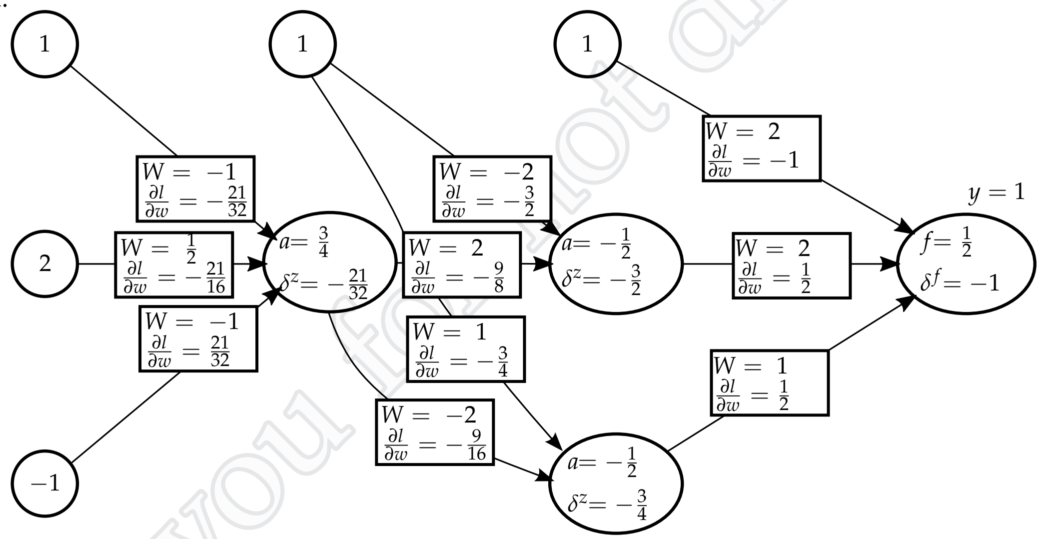
a. $\boldsymbol{W}^{(1)}$ is 7 by 3, $\boldsymbol{W}^{(2)}$ is 5 by 7, and $\boldsymbol{W}^{(3)}$ is 1 by 5.
b. $\boldsymbol{W}^{(1)}$ is 7 by 4, $\boldsymbol{W}^{(2)}$ is 5 by 8, and $\boldsymbol{W}^{(3)}$ is 1 by 6.

Answer 8.2

a. I: 5 hidden units, II: 1 hidden unit, III: 3 hidden units
b. I: $\lambda = 1$, II: $\lambda = 1000$, III: $\lambda = 100$

Answer 8.3

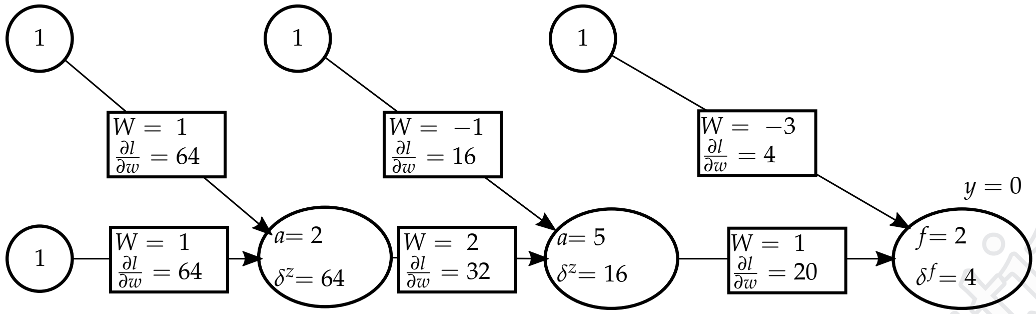
a. b. c.



d.

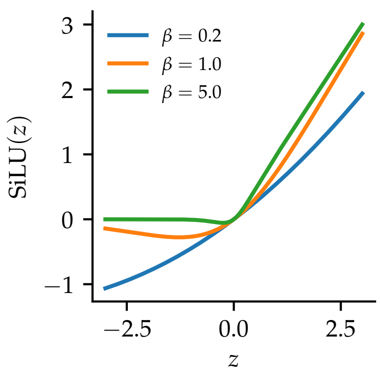
$$\begin{aligned}\boldsymbol{W}^{(1)} &= \begin{bmatrix}-1 & \frac{1}{2} & -1\end{bmatrix} - \frac{1}{3}\begin{bmatrix}-\frac{21}{32} & -\frac{21}{16} & -\frac{21}{32}\end{bmatrix} \\ &= \begin{bmatrix}-\frac{25}{32} & \frac{15}{16} & -\frac{39}{32}\end{bmatrix}\end{aligned}$$

Answer 8.4



Answer 8.5

a. Plot:



b. For small values of β, the function saturates as $z \leftarrow -\infty$ and is linear near $z = 0$.

c. The function is linear:

$$\lim_{\beta\to 0} \frac{z}{1+e^{-\beta z}} = \frac{z}{1+1} = \frac{z}{2}$$

d. The function is the ReLU:

$$\lim_{\beta\to 0} \frac{z}{1+e^{-\beta z}} = \begin{cases} z & \text{if } z > 0 \\ 0 & \text{if } z < 0 \end{cases}$$

e.

$$\begin{aligned} \frac{\partial g}{\partial z} &= \frac{\partial}{\partial z}\left(z\sigma(\beta z)\right) \\ &= \sigma(\beta z) + \beta z\sigma(\beta z)\left(1-\sigma(\beta z)\right) \\ &= g(z)\left(\frac{1}{z} + \beta - \frac{g(z)}{z}\right) \end{aligned}$$

(The middle line is fine as an answer.)

f. To do gradient descent, we first do forward propagation to get z and $a = g(z)$ (for this unit). We then do backward propagation to get $\delta^a = \frac{\partial l}{\partial a}$ (for this unit).

$$\begin{aligned}\frac{\partial l}{\partial \beta} &= \frac{\partial l}{\partial a}\frac{\partial a}{\partial \beta} \\ &= \delta^a \frac{\partial g(z)}{\partial \beta} \\ &= \delta^a z^2 \sigma(\beta z)\,(1-\sigma(\beta z)) \\ &= \delta^a g(z)\,(z - g(z)) \\ &= \delta^a a(z-a)\end{aligned}$$

Thus, for this single example, we can make the update

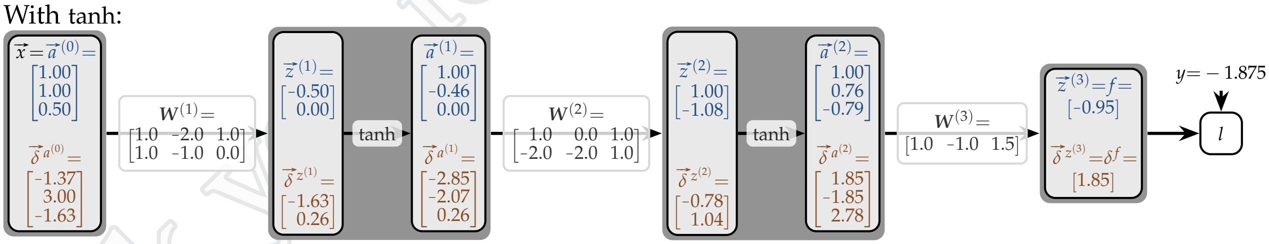
$$\beta \leftarrow \beta \underbrace{-\eta\delta^a a(z-a)}_{\text{change}}$$

Or, we can sum up the change over multiple examples as the batch update.

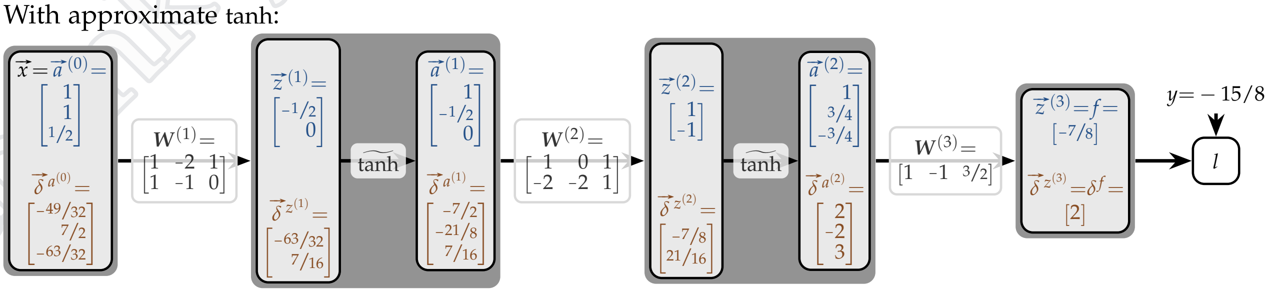
Answer 8.6

a. forward b. backward

With tanh:



With approximate tanh:



c. Resulting gradients are

With tanh:

$$\nabla_{W^{(1)}} l = \begin{bmatrix} 1.00 \\ 1.00 \\ 0.50 \end{bmatrix} \begin{bmatrix} -1.63 \\ 0.26 \end{bmatrix}^\top = \begin{bmatrix} -1.63 & -1.63 & -0.82 \\ 0.26 & 0.26 & 0.13 \end{bmatrix}$$

$$\nabla_{W^{(2)}} l = \begin{bmatrix} 1.00 \\ -0.46 \\ 0.00 \end{bmatrix} \begin{bmatrix} -0.78 \\ 1.04 \end{bmatrix}^\top = \begin{bmatrix} -0.78 & 0.36 & 0.00 \\ 1.04 & -0.48 & 0.00 \end{bmatrix}$$

$$\nabla_{W^{(3)}} l = \begin{bmatrix} 1.00 \\ 0.76 \\ -0.79 \end{bmatrix} \begin{bmatrix} 1.85 \end{bmatrix}^\top = \begin{bmatrix} 1.85 & 1.41 & -1.47 \end{bmatrix}$$

With approximate $\tanh$:

$$\nabla_{W^{(1)}} l = \begin{bmatrix} 1 \\ 1 \\ 1/2 \end{bmatrix} \begin{bmatrix} -63/32 \\ 7/16 \end{bmatrix}^\top = \begin{bmatrix} -63/32 & -63/32 & -63/64 \\ 7/16 & 7/16 & 7/32 \end{bmatrix}$$

$$\nabla_{W^{(2)}} l = \begin{bmatrix} 1 \\ -1/2 \\ 0 \end{bmatrix} \begin{bmatrix} -7/8 \\ 21/16 \end{bmatrix}^\top = \begin{bmatrix} -7/8 & 7/16 & 0 \\ 21/16 & -21/32 & 0 \end{bmatrix}$$

$$\nabla_{W^{(3)}} l = \begin{bmatrix} 1 \\ 3/4 \\ -3/4 \end{bmatrix} \begin{bmatrix} 2 \end{bmatrix}^\top = \begin{bmatrix} 2 & 3/2 & -3/2 \end{bmatrix}$$

d. We *subtract* 0.25 times the gradient from the corresponding weight.
With $\tanh$:

$$W^{(1)} l = \begin{bmatrix} 1.00 & -2.00 & 1.00 \\ 1.00 & -1.00 & 0.00 \end{bmatrix} - 0.25 \begin{bmatrix} -1.63 & -1.63 & -0.82 \\ 0.26 & 0.26 & 0.13 \end{bmatrix} = \begin{bmatrix} 1.41 & -1.59 & 1.20 \\ 0.94 & -1.06 & -0.03 \end{bmatrix}$$

$$W^{(2)} l = \begin{bmatrix} 1.00 & 0.00 & 1.00 \\ -2.00 & -2.00 & 1.00 \end{bmatrix} - 0.25 \begin{bmatrix} -0.78 & 0.36 & 0.00 \\ 1.04 & -0.48 & 0.00 \end{bmatrix} = \begin{bmatrix} 1.19 & -0.09 & 1.00 \\ -2.26 & -1.88 & 1.00 \end{bmatrix}$$

$$W^{(3)} l = \begin{bmatrix} 1.00 & -1.00 & 1.50 \end{bmatrix} - 0.25 \begin{bmatrix} 1.85 & 1.41 & -1.47 \end{bmatrix} = \begin{bmatrix} 0.54 & -1.35 & 1.87 \end{bmatrix}$$

With approximate $\tanh$:

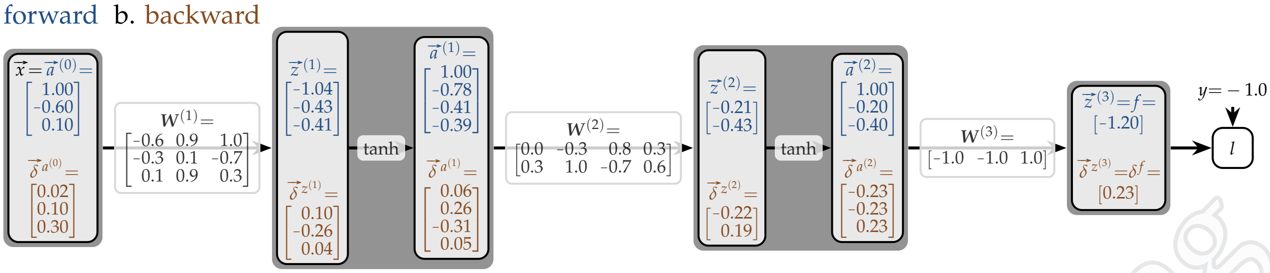
$$W^{(1)} l = \begin{bmatrix} 1 & -2 & 1 \\ 1 & -1 & 0 \end{bmatrix} - \frac{1}{4} \begin{bmatrix} -63/32 & -63/32 & -63/64 \\ 7/16 & 7/16 & 7/32 \end{bmatrix} = \begin{bmatrix} 191/128 & -193/128 & 319/256 \\ 57/64 & -71/64 & -7/128 \end{bmatrix}$$

$$W^{(2)} l = \begin{bmatrix} 1 & 0 & 1 \\ -2 & -2 & 1 \end{bmatrix} - \frac{1}{4} \begin{bmatrix} -7/8 & 7/16 & 0 \\ 21/16 & -21/32 & 0 \end{bmatrix} = \begin{bmatrix} 39/32 & -7/64 & 1 \\ -149/64 & -235/128 & 1 \end{bmatrix}$$

$$W^{(3)} l = \begin{bmatrix} 1 & -1 & 3/2 \end{bmatrix} - \frac{1}{4} \begin{bmatrix} 2 & 3/2 & -3/2 \end{bmatrix} = \begin{bmatrix} 1/2 & -11/8 & 15/8 \end{bmatrix}$$

Answer 8.7

a. forward b. backward



Yes. $f(\vec{x})$ is less than zero, so it predicts the negative class, which is the correct class for this example.

c. Resulting gradients are

$$\nabla_{W^{(1)}} l = \begin{bmatrix} 1.00 \\ -0.60 \\ 0.10 \end{bmatrix} \begin{bmatrix} 0.10 \\ -0.26 \\ 0.04 \end{bmatrix}^\top = \begin{bmatrix} 0.10 & -0.06 & 0.01 \\ -0.26 & 0.16 & -0.03 \\ 0.04 & -0.03 & 0.00 \end{bmatrix}$$

$$\nabla_{W^{(2)}} l = \begin{bmatrix} 1.00 \\ -0.78 \\ -0.41 \\ -0.39 \end{bmatrix} \begin{bmatrix} -0.22 \\ 0.19 \end{bmatrix}^\top = \begin{bmatrix} -0.22 & 0.17 & 0.09 & 0.09 \\ 0.19 & -0.15 & -0.08 & -0.08 \end{bmatrix}$$

$$\nabla_{W^{(3)}} l = \begin{bmatrix} 1.00 \\ -0.20 \\ -0.40 \end{bmatrix} \begin{bmatrix} 0.23 \end{bmatrix}^\top = \begin{bmatrix} 0.23 & -0.05 & -0.09 \end{bmatrix}$$

d. We *subtract* 0.1 times the gradient from the corresponding weight.

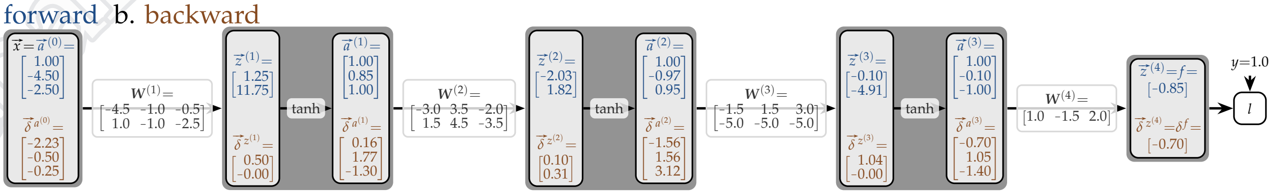
$$W^{(1)} l = \begin{bmatrix} -0.60 & 0.90 & 1.00 \\ -0.30 & 0.10 & -0.70 \\ 0.10 & 0.90 & 0.30 \end{bmatrix} - 0.1 \begin{bmatrix} 0.10 & -0.06 & 0.01 \\ -0.26 & 0.16 & -0.03 \\ 0.04 & -0.03 & 0.00 \end{bmatrix} = \begin{bmatrix} -0.61 & 0.91 & 1.00 \\ -0.27 & 0.08 & -0.70 \\ 0.10 & 0.90 & 0.30 \end{bmatrix}$$

$$W^{(2)} l = \begin{bmatrix} 0.00 & -0.30 & 0.80 & 0.30 \\ 0.30 & 1.00 & -0.70 & 0.60 \end{bmatrix} - 0.1 \begin{bmatrix} -0.22 & 0.17 & 0.09 & 0.09 \\ 0.19 & -0.15 & -0.08 & -0.08 \end{bmatrix} = \begin{bmatrix} 0.02 & -0.32 & 0.79 & 0.29 \\ 0.28 & 1.02 & -0.69 & 0.61 \end{bmatrix}$$

$$W^{(3)} l = \begin{bmatrix} -1.00 & -1.00 & 1.00 \end{bmatrix} - 0.1 \begin{bmatrix} 0.23 & -0.05 & -0.09 \end{bmatrix} = \begin{bmatrix} -1.02 & -1.00 & 1.01 \end{bmatrix}$$

Answer 8.8

a. forward b. backward



No. $f(\vec{x})$ is less than zero, so it predicts the negative class, which is the incorect class for this example.

c. Resulting gradients are

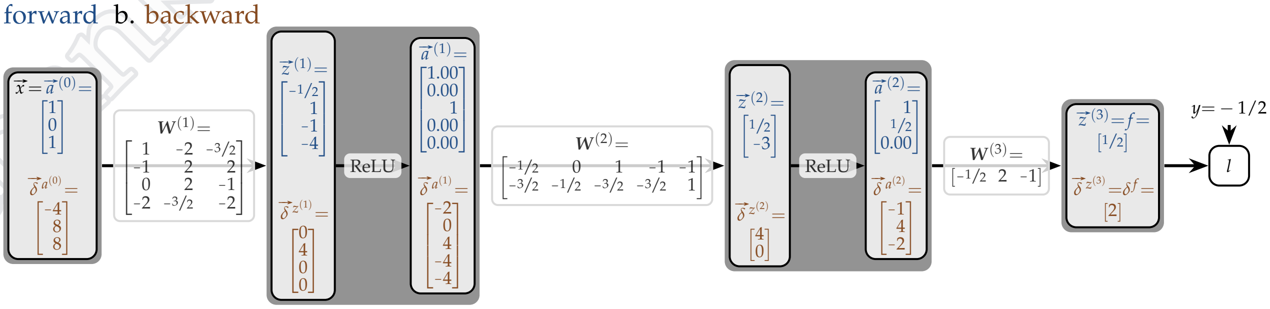
$$
\begin{aligned}
\nabla_{W^{(1)}} l &= \begin{bmatrix} 1.00 \\ -4.50 \\ -2.50 \end{bmatrix} \begin{bmatrix} 0.50 \\ -0.00 \end{bmatrix}^\top = \begin{bmatrix} 0.50 & -2.23 & -1.24 \\ -0.00 & 0.00 & 0.00 \end{bmatrix} \\
\nabla_{W^{(2)}} l &= \begin{bmatrix} 1.00 \\ 0.85 \\ 1.00 \end{bmatrix} \begin{bmatrix} 0.10 \\ 0.31 \end{bmatrix}^\top = \begin{bmatrix} 0.10 & 0.09 & 0.10 \\ 0.31 & 0.26 & 0.31 \end{bmatrix} \\
\nabla_{W^{(3)}} l &= \begin{bmatrix} 1.00 \\ -0.97 \\ 0.95 \end{bmatrix} \begin{bmatrix} 1.04 \\ -0.00 \end{bmatrix}^\top = \begin{bmatrix} 1.04 & -1.00 & 0.98 \\ -0.00 & 0.00 & -0.00 \end{bmatrix} \\
\nabla_{W^{(4)}} l &= \begin{bmatrix} 1.00 \\ -0.10 \\ -1.00 \end{bmatrix} \begin{bmatrix} -0.70 \end{bmatrix}^\top = \begin{bmatrix} -0.70 & 0.07 & 0.70 \end{bmatrix}
\end{aligned}
$$

d. We *subtract* 0.1 times the gradient from the corresponding weight.

$$
\begin{aligned}
W^{(1)} l &= \begin{bmatrix} -4.50 & -1.00 & -0.50 \\ 1.00 & -1.00 & -2.50 \end{bmatrix} - 0.1 \begin{bmatrix} 0.50 & -2.23 & -1.24 \\ -0.00 & 0.00 & 0.00 \end{bmatrix} = \begin{bmatrix} -4.55 & -0.78 & -0.38 \\ 1.00 & -1.00 & -2.50 \end{bmatrix} \\
W^{(2)} l &= \begin{bmatrix} -3.00 & 3.50 & -2.00 \\ 1.50 & 4.50 & -3.50 \end{bmatrix} - 0.1 \begin{bmatrix} 0.10 & 0.09 & 0.10 \\ 0.31 & 0.26 & 0.31 \end{bmatrix} = \begin{bmatrix} -3.01 & 3.49 & -2.01 \\ 1.47 & 4.47 & -3.53 \end{bmatrix} \\
W^{(3)} l &= \begin{bmatrix} -1.50 & 1.50 & 3.00 \\ -5.00 & -5.00 & -5.00 \end{bmatrix} - 0.1 \begin{bmatrix} 1.04 & -1.00 & 0.98 \\ -0.00 & 0.00 & -0.00 \end{bmatrix} = \begin{bmatrix} -1.60 & 1.60 & 2.90 \\ -5.00 & -5.00 & -5.00 \end{bmatrix} \\
W^{(4)} l &= \begin{bmatrix} 1.00 & -1.50 & 2.00 \end{bmatrix} - 0.1 \begin{bmatrix} -0.70 & 0.07 & 0.70 \end{bmatrix} = \begin{bmatrix} 1.07 & -1.51 & 1.93 \end{bmatrix}
\end{aligned}
$$

Answer 8.9

a. forward b. backward



c. Resulting gradients are

$$\nabla_{\boldsymbol{W}^{(1)}} l = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \\ 4 \\ 0 \\ 0 \end{bmatrix}^\top = \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\nabla_{\boldsymbol{W}^{(2)}} l = \begin{bmatrix} 1.00 \\ 0.00 \\ 1 \\ 0.00 \\ 0.00 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \end{bmatrix}^\top = \begin{bmatrix} 4 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\nabla_{\boldsymbol{W}^{(3)}} l = \begin{bmatrix} 1 \\ 1/2 \\ 0.00 \end{bmatrix} \begin{bmatrix} 2 \end{bmatrix}^\top = \begin{bmatrix} 2 & 1 & 0 \end{bmatrix}$$

d. We *subtract* $\frac{1}{4}$ times the gradient from the corresponding weight.
With approximate tanh:

$$\boldsymbol{W}^{(1)} l = \begin{bmatrix} 1 & -2 & -3/2 \\ -1 & 2 & 2 \\ 0 & 2 & -1 \\ -2 & -3/2 & -2 \end{bmatrix} - \frac{1}{4} \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -2 & -3/2 \\ -2 & 2 & 1 \\ 0 & 2 & -1 \\ -2 & -3/2 & -2 \end{bmatrix}$$

$$\boldsymbol{W}^{(2)} l = \begin{bmatrix} -1/2 & 0 & 1 & -1 & -1 \\ -3/2 & -1/2 & -3/2 & -3/2 & 1 \end{bmatrix} - \frac{1}{4} \begin{bmatrix} 4 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} -3/2 & 0 & 0 & -1 & -1 \\ -3/2 & -1/2 & -3/2 & -3/2 & 1 \end{bmatrix}$$

$$\boldsymbol{W}^{(3)} l = \begin{bmatrix} -1/2 & 2 & -1 \end{bmatrix} - \frac{1}{4} \begin{bmatrix} 2 & 1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 7/4 & -1 \end{bmatrix}$$